Design a Stockbroker

SystemsExpert design walkthrough

Clarifying questions

- Do we run the exchange or route orders to one?
- Order types (market, limit), and is trading only during market hours?
- How many users and orders per day?
- Need real-time price feeds in the app?

Functional requirements

- Place, cancel and view orders
- See balances and positions
- Order status updates when filled

Non-functional requirements

- Never lose or duplicate an order; balances must be correct
- Low latency to the exchange
- Highly available during market hours

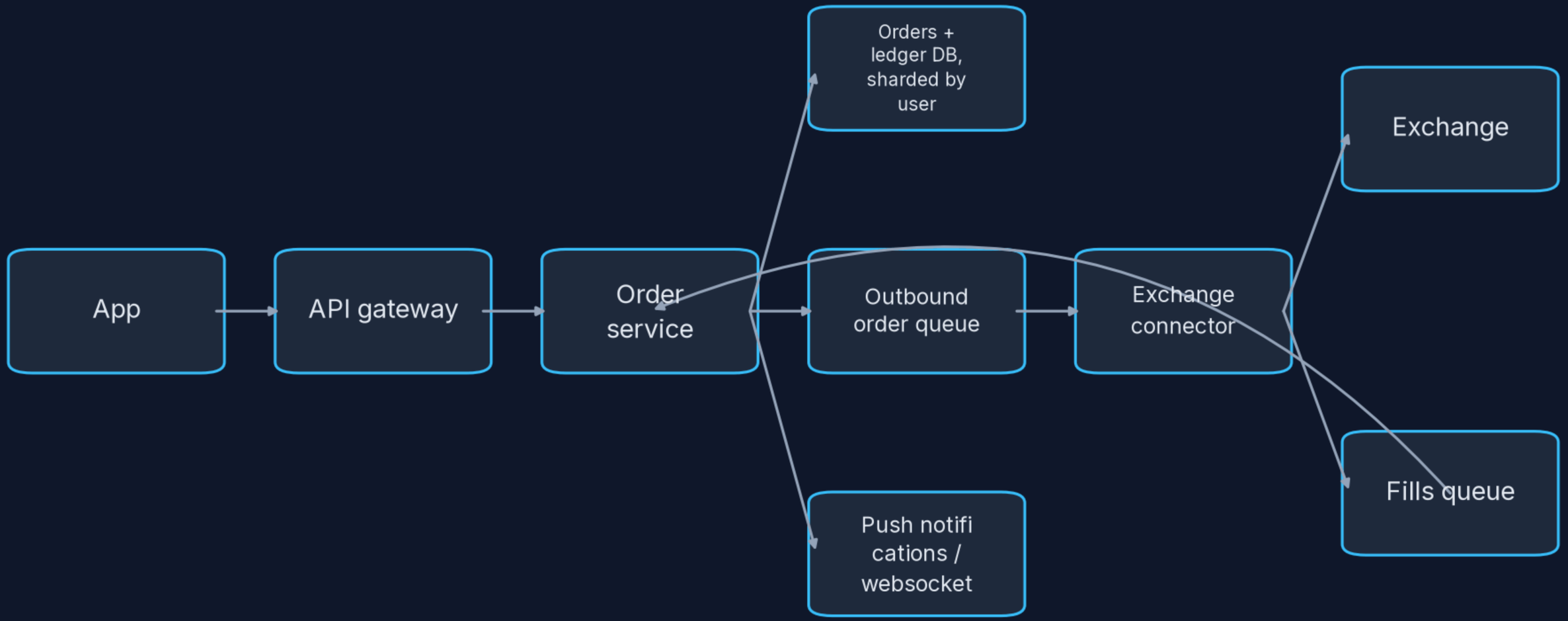
Capacity estimates

- ~10M users, ~20M orders/day, peaks at open/close ~2k orders/s
- Each order ~1 KB → tiny storage; correctness dominates design

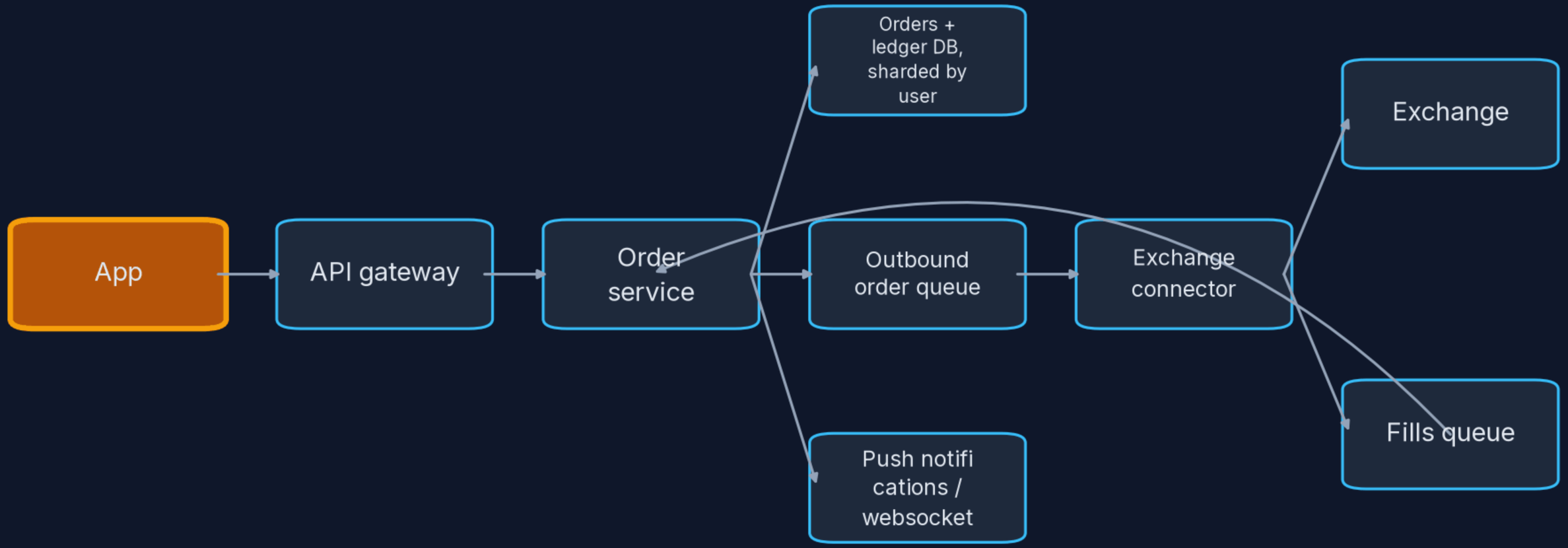
Capacity: worked out

- 20M orders/day ÷ 86,400 s ≈ 230 orders/s
- Market open/close spikes ~10× → ~2k orders/s
- 1 KB/order × 20M ≈ 20 GB/day
- Storage is small; correctness dominates

High-level architecture

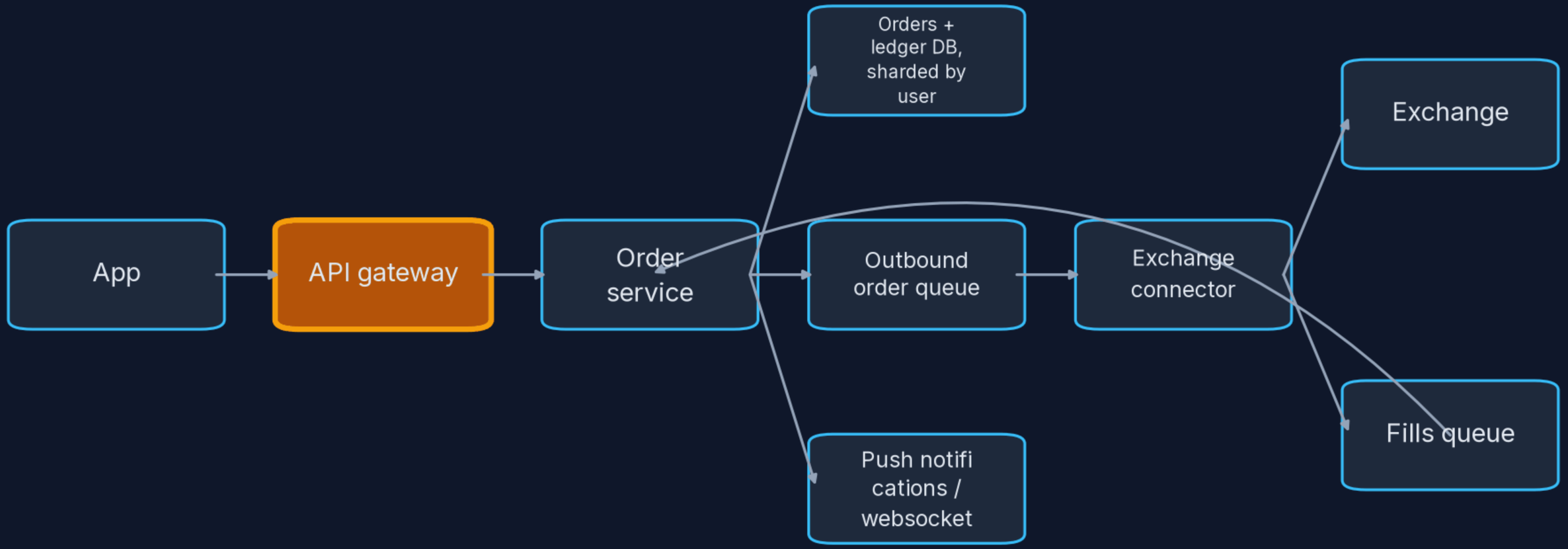


Request flow: Place a buy order



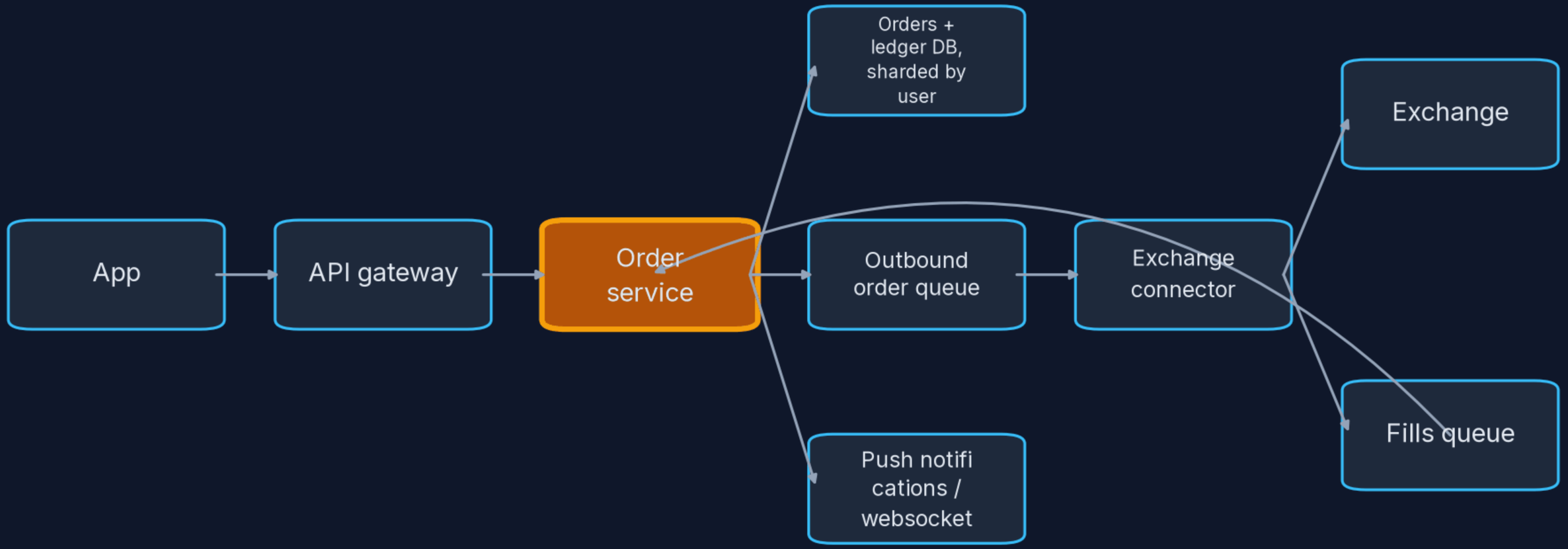
1. User taps Buy

Request flow: Place a buy order



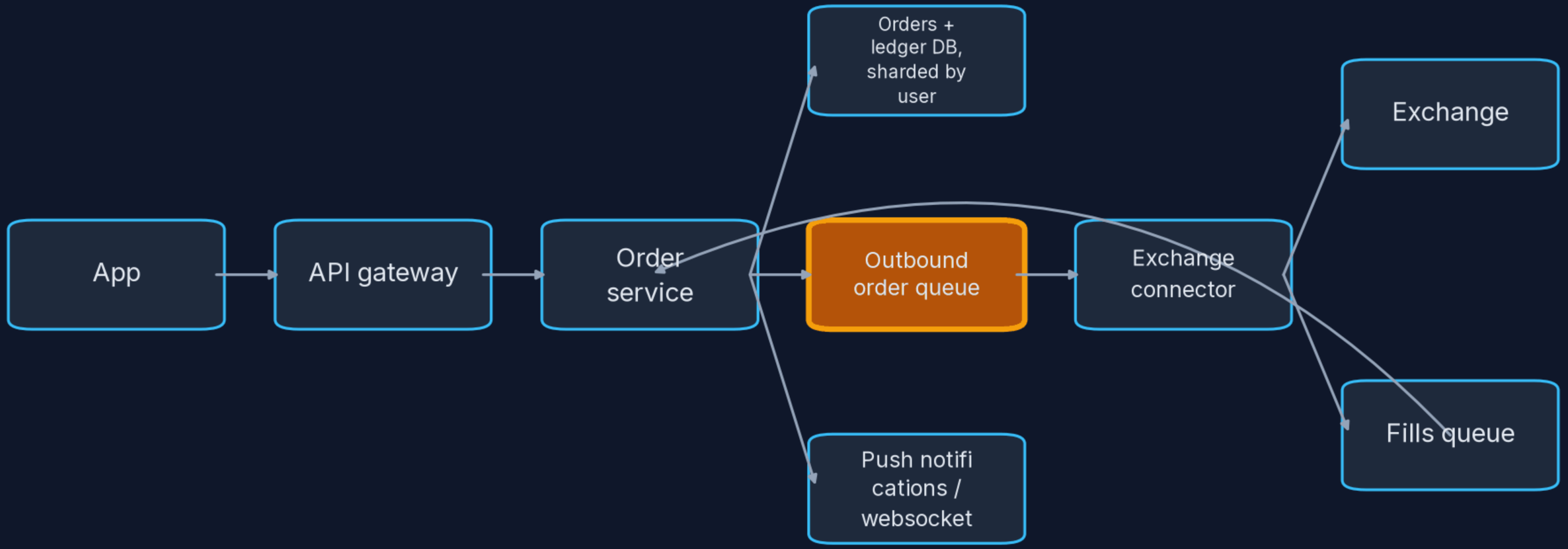
2. Gateway authenticates

Request flow: Place a buy order



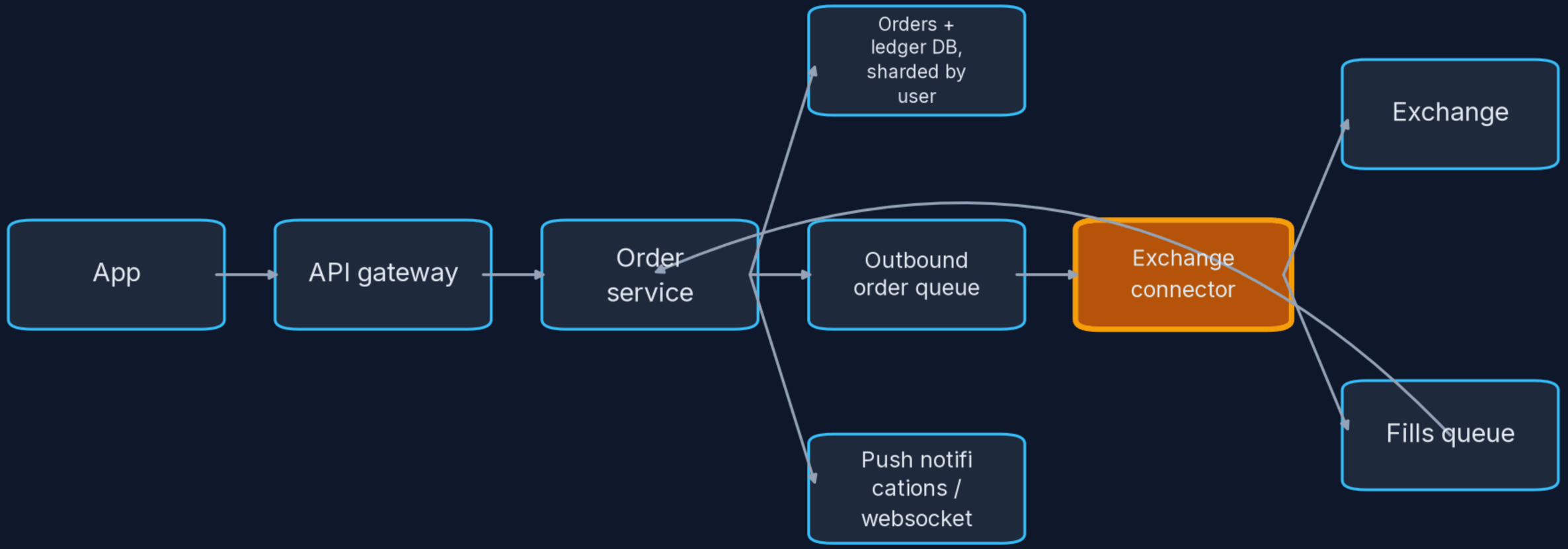
3. Check and reserve funds

Request flow: Place a buy order



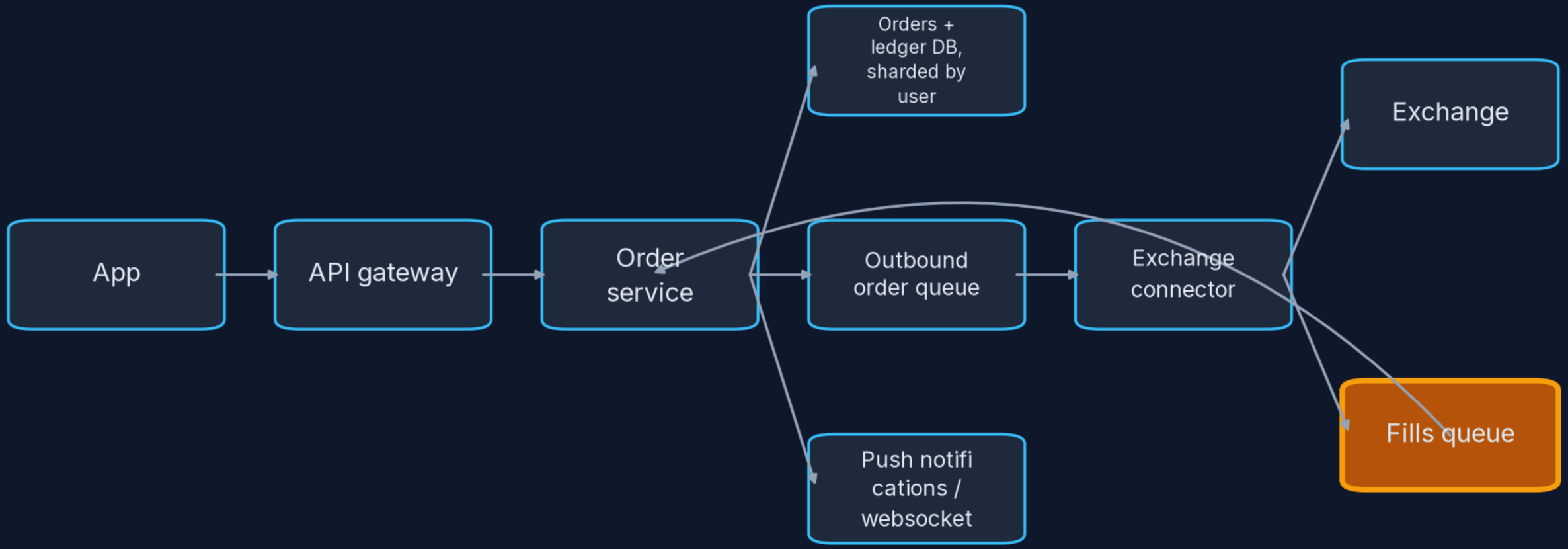
4. Order queued for the exchange

Request flow: Place a buy order



5. Connector sends to the exchange

Request flow: Place a buy order



6. Fills update the ledger

Data model

- Orders(id, user_id, idempotency_key, symbol, side, type, qty, limit_price, status, exchange_order_id)
- Ledger_entries(id, user_id, amount, currency, reason, order_id)
  - append-only, balance = sum
- Positions(user_id, symbol, qty) derived from fills

API

```
POST   /orders  {idempotencyKey, symbol, side, qty, type, limitPrice?} -> order
DELETE /orders/{id}
GET    /orders?status=open
GET    /portfolio -> balances, positions
```

Exactly-once order placement

- Client sends an idempotency key; order row is written (status PENDING) in the same transaction that reserves buying power.
- A connector picks PENDING orders, sends them with our client order id so retries are deduplicated by the exchange.

Fills

- The exchange streams executions; connector writes them to a durable queue; order service applies them idempotently (keyed by execution id) updating order status, ledger and positions in one transaction.

Scaling

- Shard by user_id; exchange connectors partitioned by symbol range; keep hot reads (portfolio) in a cache invalidated by fills.

Key decisions

- Ledger with double-entry → auditable, never lose money
- Idempotency keys → safe client retries
- Queues around the exchange → decouple from exchange outages
- Shard by user → keeps a user's orders in one place

Trade-offs & alternatives

- Strong consistency on money paths over availability
- Queue between us and the exchange decouples bursts but adds latency (ms)
- Ledger vs mutable balance: ledger is auditable; cache the sum

Recap

- Clarify scope, then functional + non-functional needs
- Estimate load to size storage and fan-out
- Core pieces: App, API gateway, Order service, Orders + ledger DB, sharded by user, Outbound order queue
- Know your trade-offs and failure modes